

ServiceNow - Discovery



Discovery finds computers, servers, printers, a variety of IP-enabled devices, and the applications that run on them. It can then update the CIs in your CMDB with the data it collects. **Plugin : sn itom pattern**

There are two types of discovery:

- **Horizontal discovery**
- **Top-down discovery**



Horizontal discovery is a technique Discovery uses to scan your network, find computers and devices, and then populate the CMDB with the CIs it finds.

Horizontal discovery creates direct relationships between CIs, such as a runs-on relationship between an application CI and the actual computer CI it runs on.

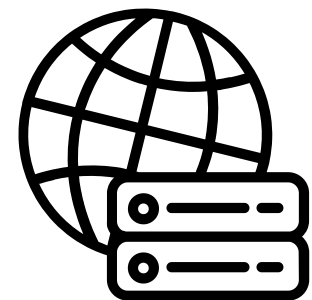
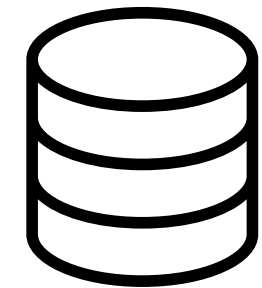
Horizontal discovery is not aware of business services and does not create relationships between CIs based on the business service they are in.

In **TopDown discovery/ServiceMapping** we provide entry point (URL) then Discovery checks it and build the business service from its CIs and connections to it. Service Mapping uses results of horizontal discovery performed by Discovery.

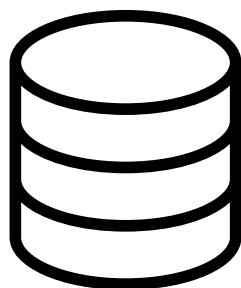


ServiceNow - Discovery

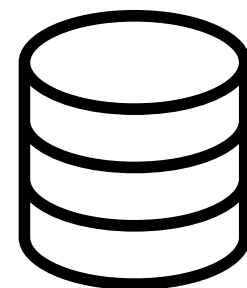
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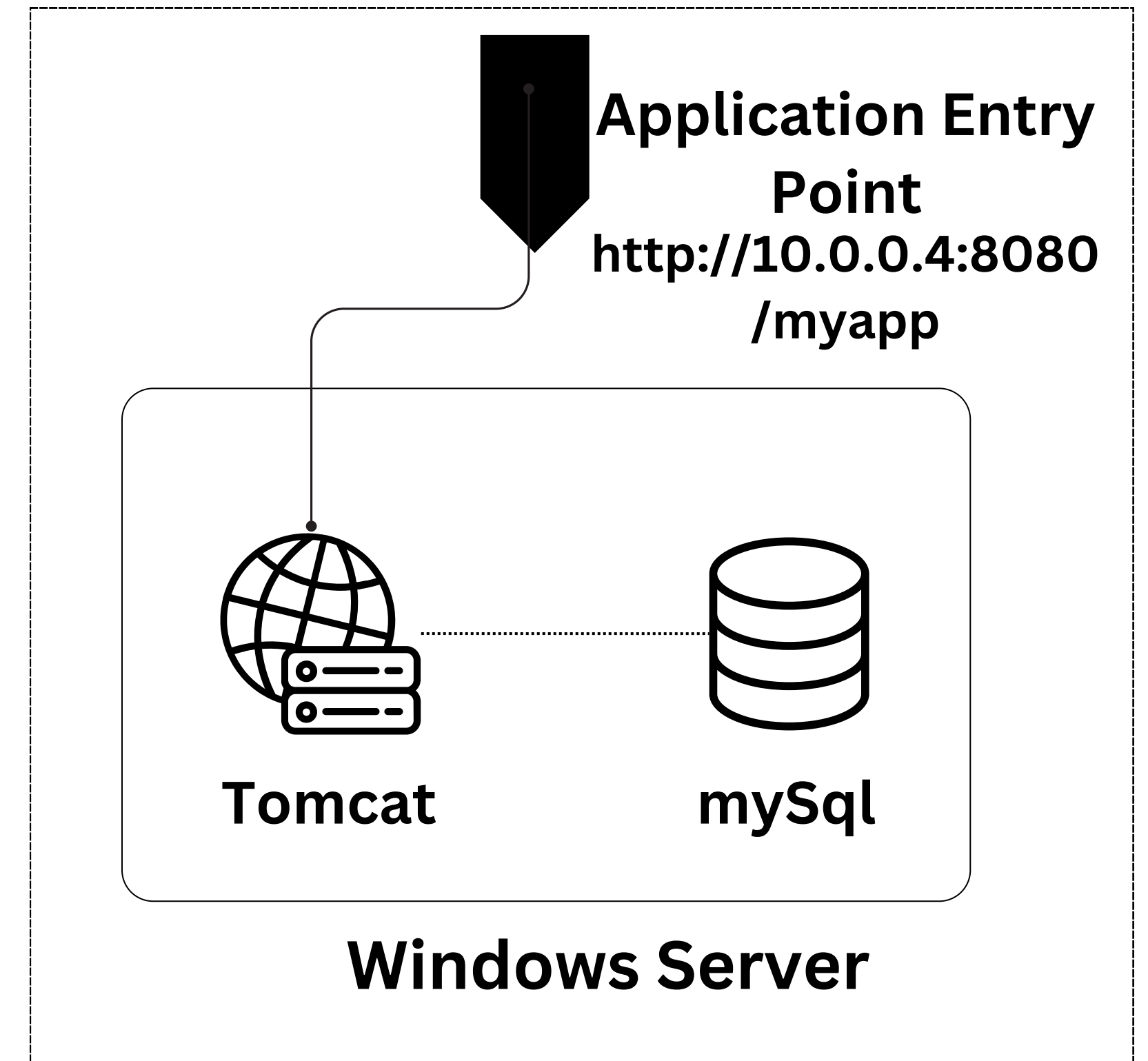
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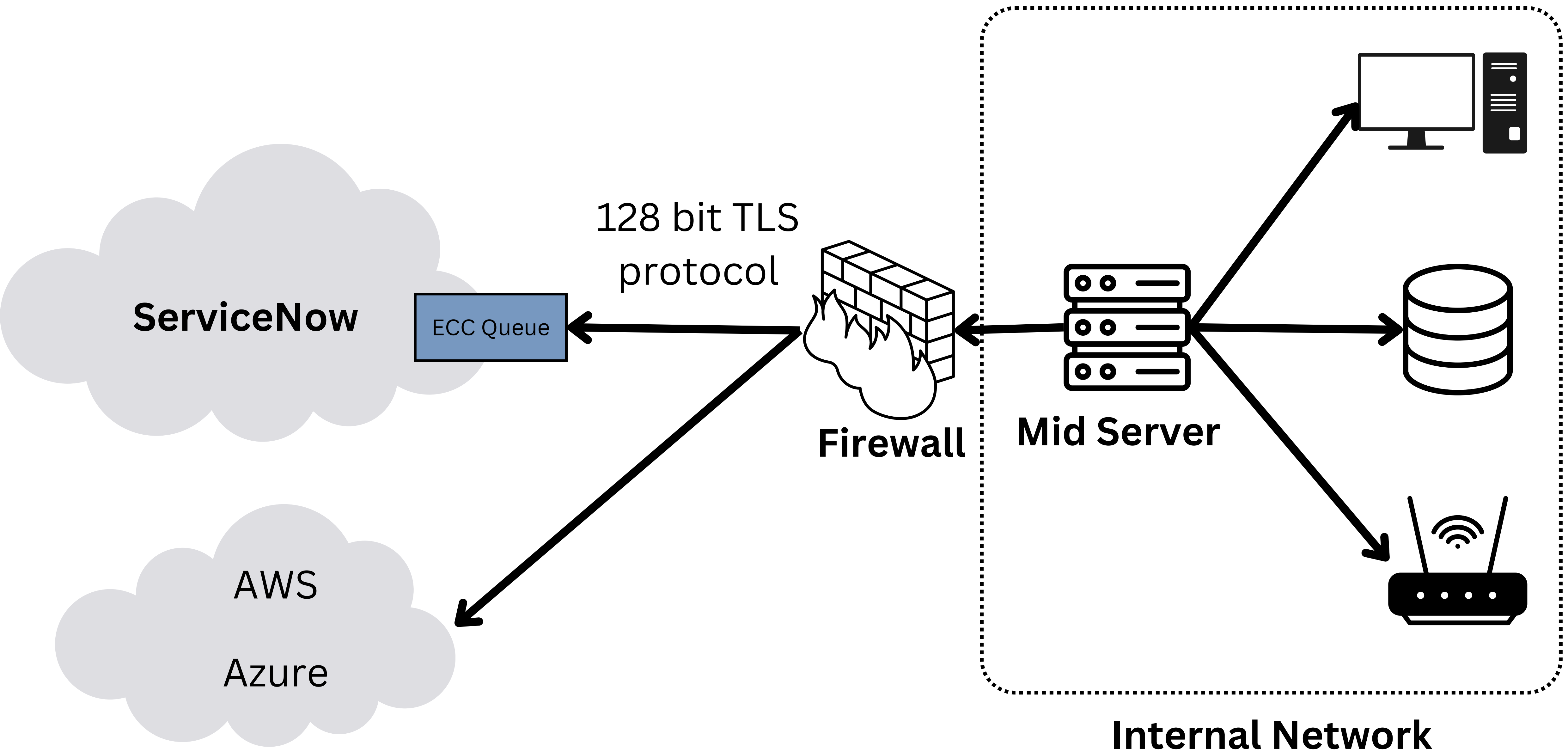
Horizontal Discovery



Top Down Discovery

Follow on





Public Cloud

Follow on



Discovery Components

Discovery uses these components to find CIs:

- **Probes and Sensors** - Written in JavaScript. In Discovery the job of Probe is to collect the information and then Sensor processes that information to update the CMDB. Both get their instructions from the ECC queue.
- **Patterns** - Written in Neebula Discovery Language (NDL). A Pattern is a series of operations that collect data, process it, and update the CMDB. A pattern performs the same function as a probe.



Probes

Runs during all discovery phases.

Supports all standard network, and CI discovery.

Creating or customizing a probe requires knowledge of JAVA, JAVA script

In identification or exploration phase, Probe can take longer time for discovery because of sending multiple probes.

Patterns

Runs during the identification and exploration phases only.

Supports all standard network and ci discovery.
Supports Cloud discovery, configuration file tracking

Pattern designer is a user friendly interface to create or customize a pattern

Pattern has faster performance as all the operations needed to execute are inside the pattern only.



Horizontal discovery phases

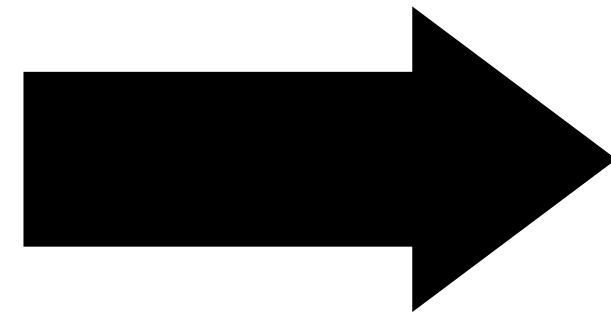
Discovery uses these components to find CIs:

- **Scanning** - Sends Shazzam probe to check open ports in network (eg: 135 for windows server)
- **Classification** - Checks for type of device or OS on device
- **Identification** - Create or update CI in CMDB, Use IRE rules
- **Exploration** - Gather additional information on devices ie : applications, memory etc and create relationship



Start Discovery

IP - 10.0.0.6



Scanning

Shazzam Prob scan for active port in device IP 10.0.0.6

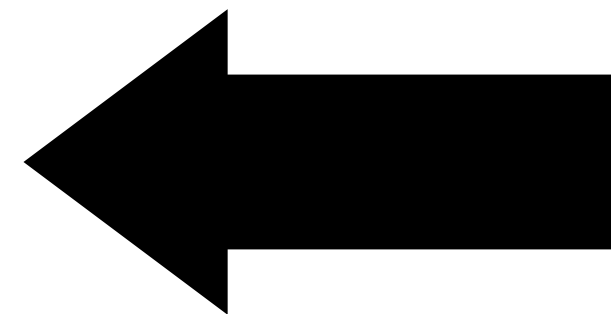
Classification

Port 135 is active (Windows Device), Sends probe “Windows - Classify”

Identification

Execute Patterns “Windows OS - Servers” for Identification and Exploration

Exploration



End Discovery

Discovery - Credentials

Mid Server uses credentials configured in **discovery_credentials** table to access resources for Discovery and Service Mapping.

Create and Test Credential

- Navigate to Discovery > Credentials
- Click New
- Click a link for the credential type and complete the form.
- Under Related Links, click Test credential.



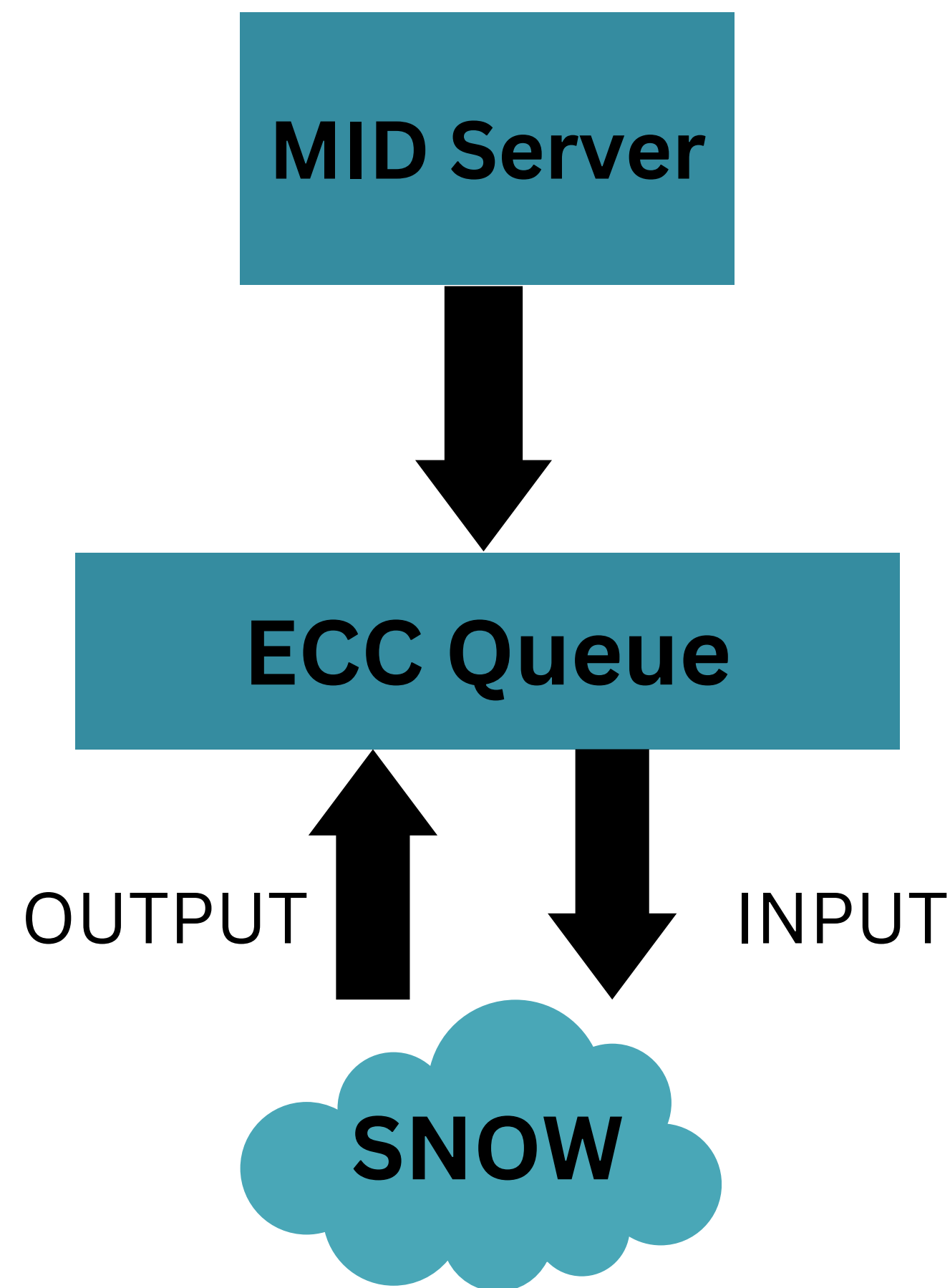
Discovery - Status

- The Discovery status provides a summary of a Discovery launched.
- Navigate to **Discovery > Status** and open the status record for a Discovery
- Discovery status displays high level information such as the date of the Discovery, the mode, the number of probe messages sent to devices, and the number of sensor records that were processed.
- Helps in troubleshoot a failed discovery.
- You can manually cancel a discovery while it is in progress from the Discovery Status record.
- A Discovery timeline generates a graphical display of a Discovery Status record
- The **Devices** related list in the Discovery Status form provides a summary list of all the devices scanned.



Discovery - ECC Queue

External Communication Channel



- Connection point between an instance and MID Server.
- Contains records that are created during all stages of discovery.
- Output queue indicates message from instance to MID server.
- Input queue indicates message from MID server to Instance.
- Provides actual XML payload that is sent to or from an instance.
- By Default records in the ECC Queue older than 7 days are deleted automatically.



Discovery - ECC Queue

External Communication Channel

ECC queue States

- When new probe instructions are available for the MID Server, the system creates an output record with the state of **ready**.
- The MID Server then starts processing the probe instructions and the state changes to **processing** before it finally changes to **processed**.
- When a result returns to the MID Server and is ready to be sent to the instance, the system creates an input record with the state of **ready**.
- While the system processes the payload, the state changes to **processing** before it finally changes to **processed**.
- If any errors are encountered, the state changes to **error**. You can look for errors to troubleshoot failed discovery.



Discovery - Schedule

What to discover, when to discover, who(MID Server) will discover.

- Run Discovery on a single IP address.
- Run a Discovery schedule manually.
- Configure the use of multiple MID Servers for load balancing.
- Configure the use of multiple Shazzam probes for load balancing.
- Create or disable a schedule that controls when the Discovery runs in your network.

Quick Discovery - Allows an administrator to run a CI Configuration discovery on a single IP address without requiring a schedule.



Discovery - Schedule Form

Name - Unique Name for schedule.

Discover - Discovery Scan types

- **Configuration items** - Uses Discovery identifiers to match devices with CIs in the CMDB and update the CMDB.
- **IP addresses** - Discover all the active IP addresses in the specified range and create device history records, but do not update the CMDB.
- **Networks** - Discovers IP networks (routers and switches).
- **Service** - Discovers services for the Service Mapping application.
- **Serverless** - Finds CIs without needing to run discovery on a host.
- **Cloud application** - Discovers only the cloud resources for the patterns that you specify.
- **Cloud resources** - Discovers resources for one of the supported cloud providers.



Discovery - Schedule Form

MID Server selection method

- **Auto-Select MID Server** - Select the MID Server automatically based on the Discovery IP Ranges you configure.
- **Specific MID Cluster** - Use a preconfigured cluster of MID Servers.
- **Specific MID Server** - Use only one MID Server.
- **Use Behavior** - Servers to perform load balancing, scanning multiple domain, scan multiple windows credentials, execute multiple protocols(SNMP, WMI or SSH) using more than one MID server.

MID Server - Selection for MID Server.

MID Server Cluster - Selection for MID Server Cluster.

Behavior - Selection for Behavior.

Location - Location to assign to the CIs that the schedule discovers.

Max run time - Set a time limit for running this schedule.



Discovery - Schedule Form

Run - Set schedule time for Discovery to run.

Log state changes - Check box to create a log entry every time the state changes during a discovery, such as a device going from Active to Classifying.

Location - Location to assign to the CIs that the schedule discovers.

Max run time - Set a time limit for running this schedule.

Shazzam batch size - Set the number of IP addresses that each Shazzam probe can scan.

Shazzam cluster support - Check box for distributing Shazzam processing among multiple MID Servers in a cluster and improve performance.

Use SNMP Version - Designate the SNMP version to use for this discovery.

Quick ranges - Create IP addresses and address ranges for schedule.

Discover now - Use this link to immediately start this Discovery.



Discovery - Schedule Related List

- **IP Ranges** - Defines the ranges of IP addresses to scan with this schedule.
- **Discovery Range Sets** - Defines each range set in a schedule to scan by one or more Shazzam probes.
- **Discovery Status** - Displays the results of current and past schedule runs.
- **Global IP exclude** - allows administrators to define global IP exclusions lists that work across Discovery schedules